

THE FUTURE IS FASTER THAN YOU THINK

How Converging Technologies Are Transforming
Business, Industries, and Our Lives

Based on the book by Peter H. Diamandis and Steven Kotler

THE BIOLOGICAL GLITCH

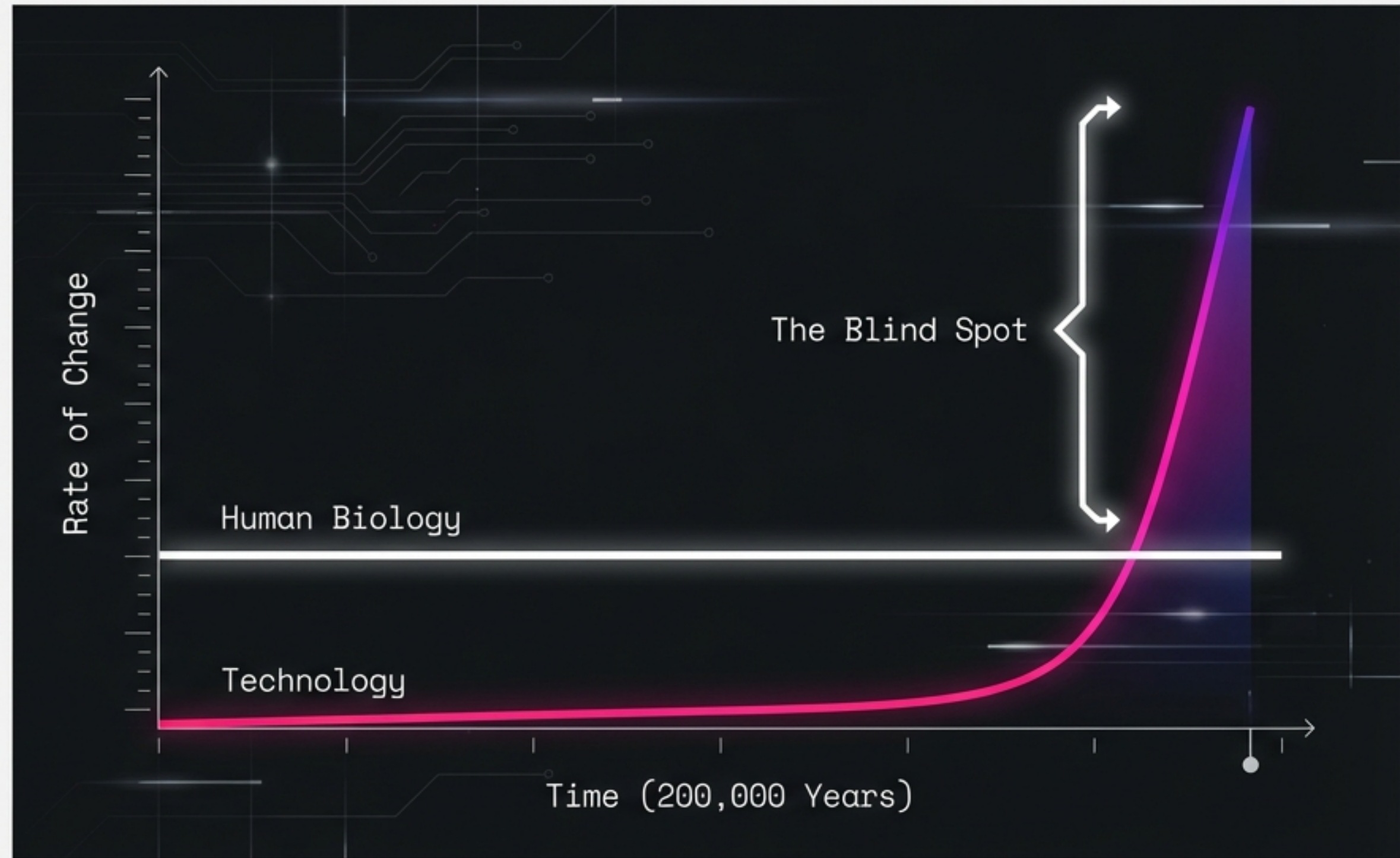
Our brains are Local and Linear.
The world is Global and Exponential.

The Problem:

The human brain hasn't had a hardware update in 200,000 years. We evolved to track local, linear change where tomorrow looked like yesterday.

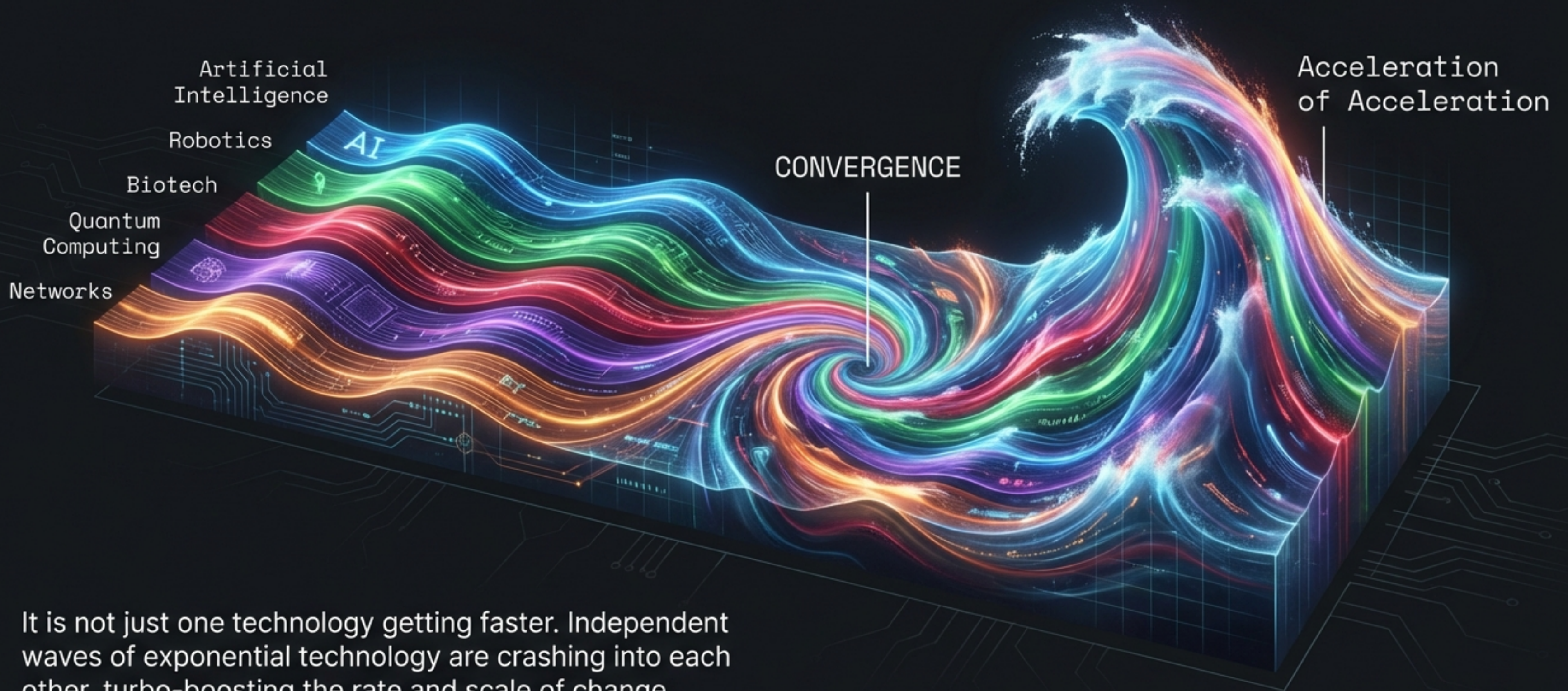
The Consequence:

We are biologically blind to what is around the bend. We struggle to process "blitzkrieg" speeds of development.



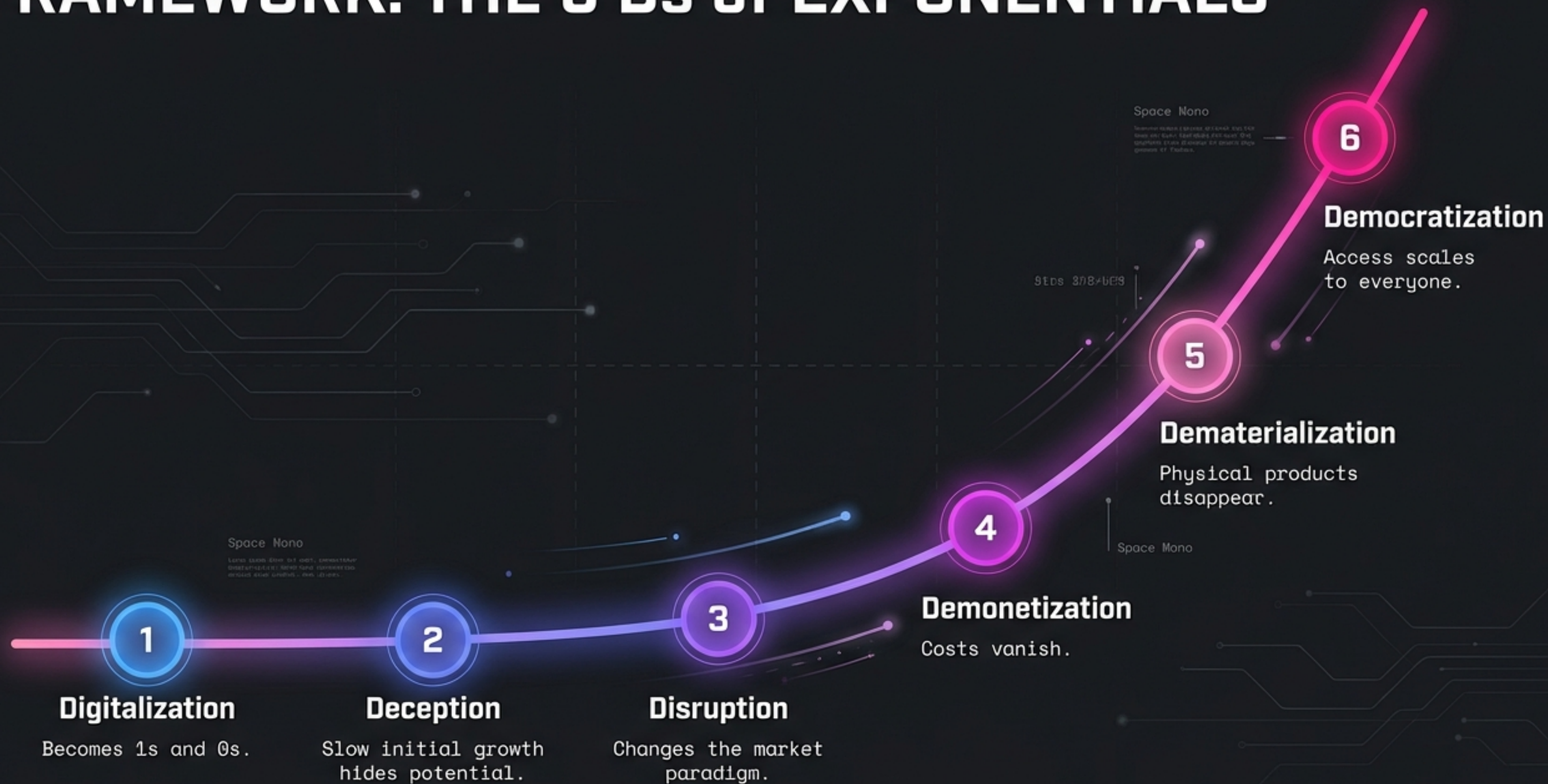
fMRI studies show the medial prefrontal cortex shuts down when we think about the future—the brain treats our future self as a stranger.

THE MECHANISM: CONVERGENCE



It is not just one technology getting faster. Independent waves of exponential technology are crashing into each other, turbo-boosting the rate and scale of change.

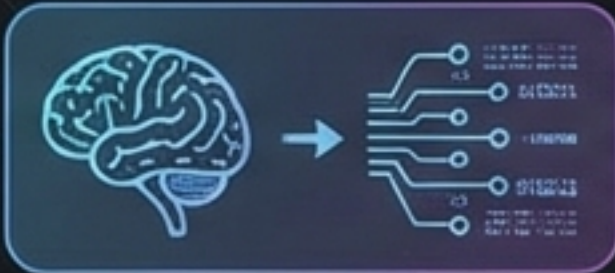
FRAMEWORK: THE 6 Ds of EXPONENTIALS



CASE STUDY: THE TRANSPORTATION REVOLUTION

Why Now?

MACHINE LEARNING

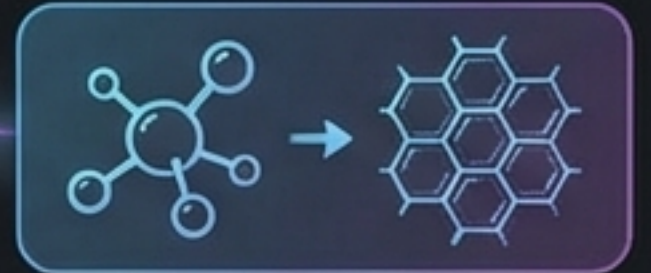


Runs complex flight simulations.

CONVERGENCE STACK

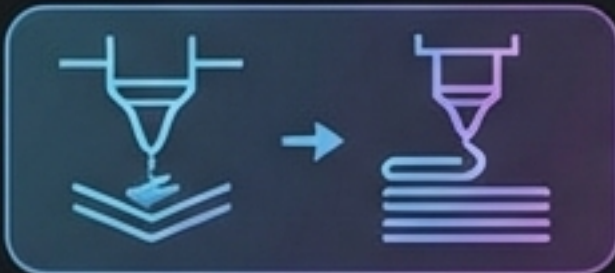


MATERIAL SCIENCE



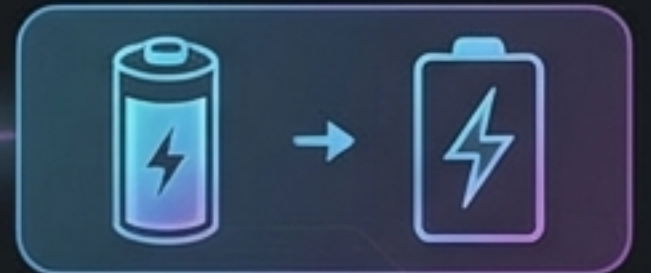
Lightweight, durable composites.

3-D PRINTING



Manufactures motors at scale.

NEW BATTERIES



High power density (350 kWh/kg).

We were promised flying cars for decades. They failed because combustion engines were too heavy. Convergence of these four technologies created Distributed Electric Propulsion (DEP)—quiet, safe, and efficient.

AERIAL RIDESHARING



COST PER PASSENGER MILE



“Ultimately, we want to make it economically irrational to own and use a car.” — Jeff Holden, Uber

AUTONOMOUS VEHICLES: THE END OF OWNERSHIP

Why Now?

CURRENT CITYSCAPE: DOMINATED BY PARKING

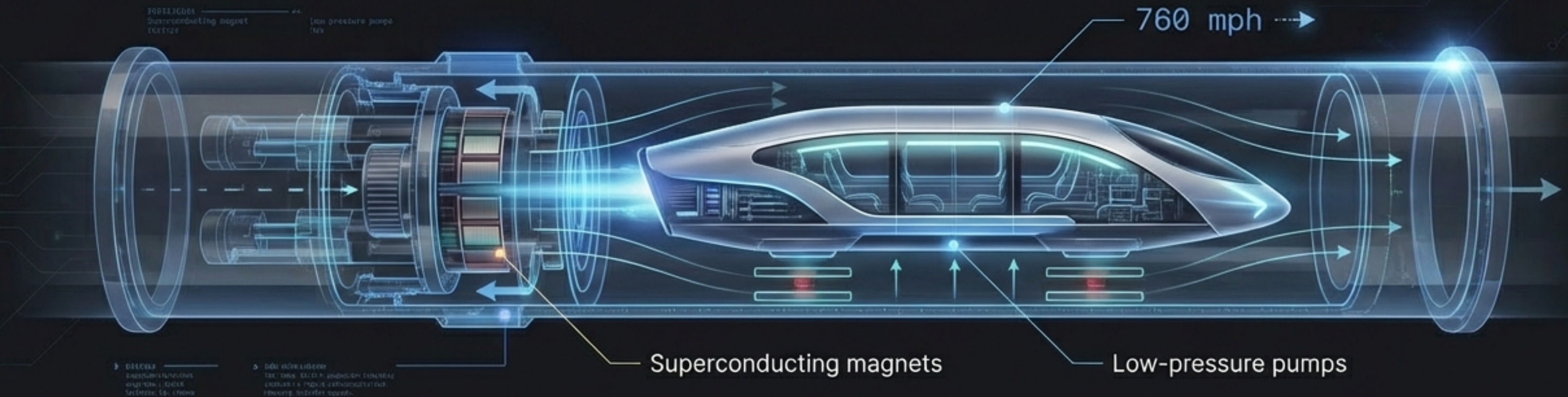


FUTURE CITYSCAPE: RECLAIMED FOR PEOPLE



Within ten years, you'll probably need a special permit to drive a human-operated car.
Ridesharing autonomous cars will be 80% cheaper than ownership.

HYPERLOOP: VACUUM TRAVEL



Los Angeles to San Francisco

35 Minutes



Chicago to DC

35 Minutes



Pune to Mumbai

25 Minutes



Side Note



- The Boring Company: Digging tunnels compatible with Hyperloop systems.
- Timeline: Certification targeted by 2023.

EARTH-TO-EARTH: ANYWHERE IN AN HOUR

NYC — Shanghai: 39 mins

London – Dubai: 29 mins

Hong Kong –
Singapore: 22 mins

Technology: SpaceX Starship

Speed:
17,500 mph

Promise:
Displacing long-haul flights with sub-orbital rocket trips. Anywhere on Earth in under an hour.

AVATARS: TRAVEL WITHOUT MOVING

VR HEADSET: SENSORY INPUT

SECURE DATA STREAM:
LOW LATENCY

ROBOTIC AVATAR: REMOTE PRESENCE



Concept: Teleporting your senses, not your body.

The Tech: VR + Haptics + Robotics.

Use Case: The ANA Avatar XPRIZE (\$10M). Occupy a robot in a distant city to walk, hold hands, and interact.

Impact: Disruption of the airline industry without moving a single body.

THE ENGINES: COMPUTING

The End of Moore's Law?

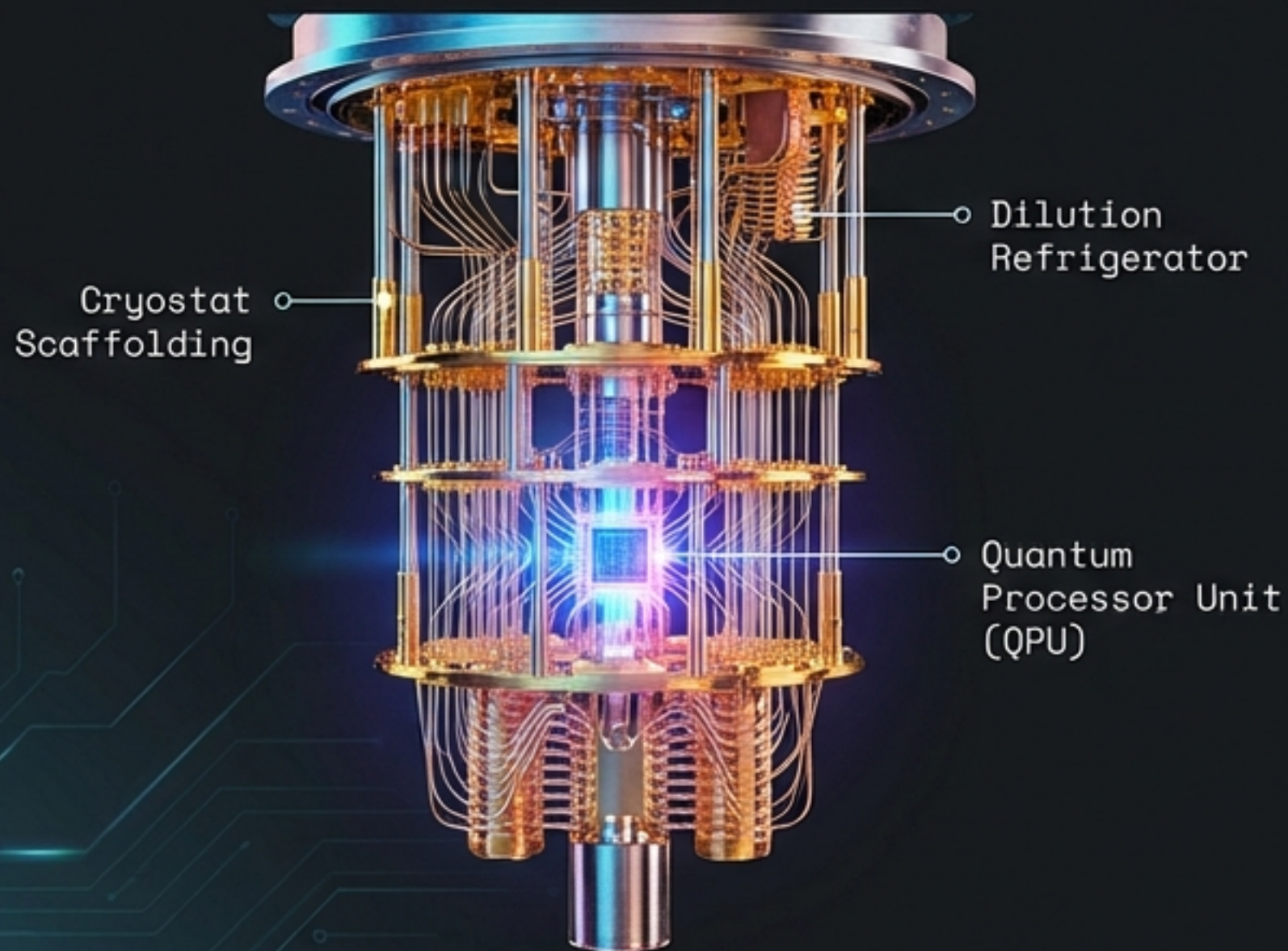
The Limit: At 5-nanometer scales, electrons jump barriers (quantum tunneling), stopping calculation

The Shift: 5th Paradigm (Integrated Circuits) -> 6th Paradigm (Quantum & Specialized Chips)

Solution: 3D Circuits, Carbon Nanotubes, and Quantum Computing

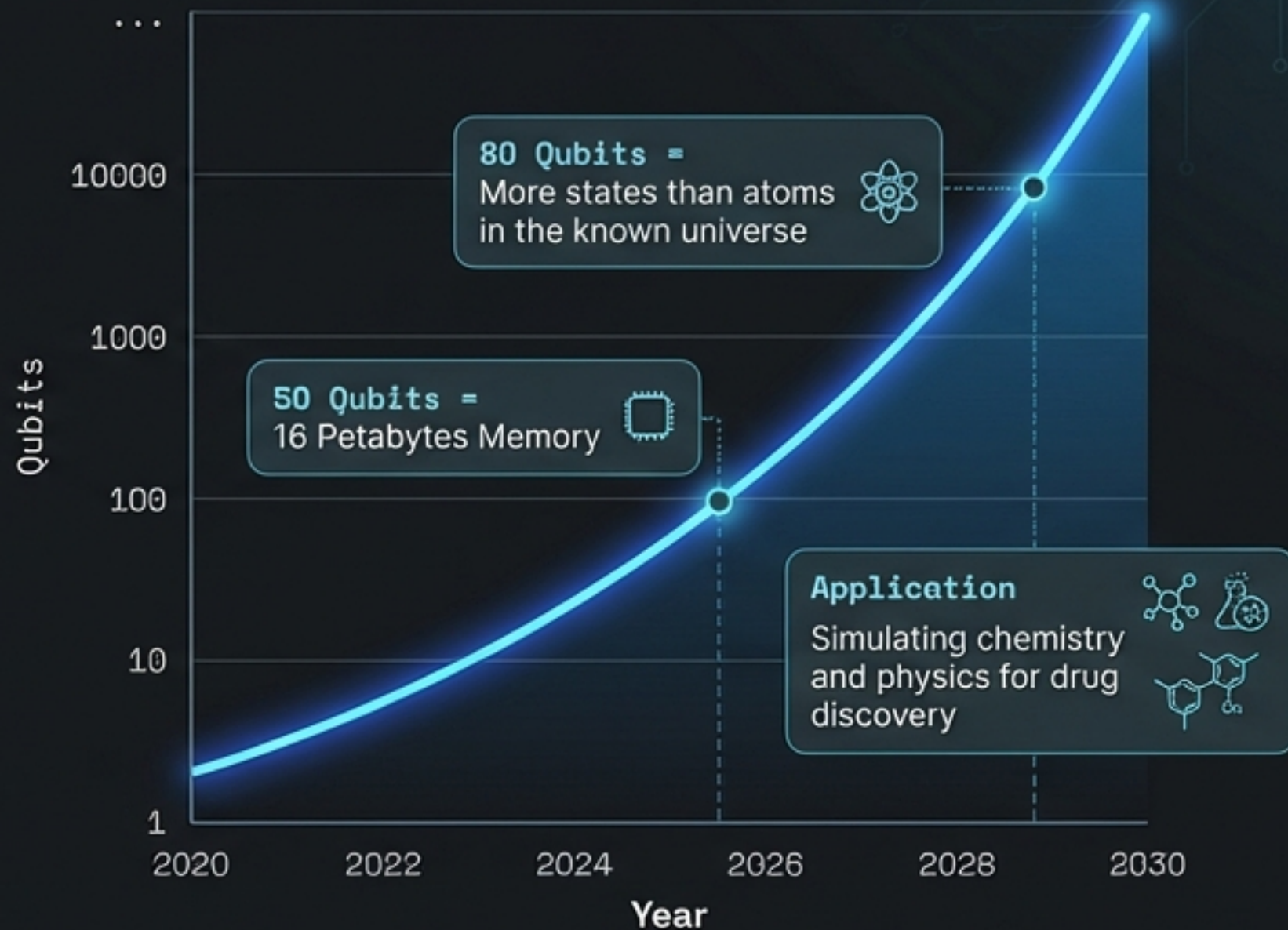
QUANTUM COMPUTING: ROSE'S LAW

The Exponential Growth of Qubits



Coldest Pipe in the Universe

ROSE'S LAW



ARTIFICIAL INTELLIGENCE: THE INTERFACE



Space Mono Bold

CASE STUDY: XIAOICE (MICROSOFT CHINA)

- Optimized for emotional connection (EQ), not just IQ.
- 30 Billion conversations.
- 100 Million human users.
- Average user interacts 60 times/month.

Space Mono Bold

2015 INFLECTION POINT:

The convergence of **Big Data** + **Cheap GPUs** pushed AI from **Deceptive** to **Disruptive**.



THE 5 SENSES OF AI



LOOKING:

Image recognition
accuracy 99.46%
(Better than human).



LISTENING:

Google Duplex
passing the Turing
test on phone calls.



READING:

Analyzing 120,000
books in 0.5 seconds.



WRITING:

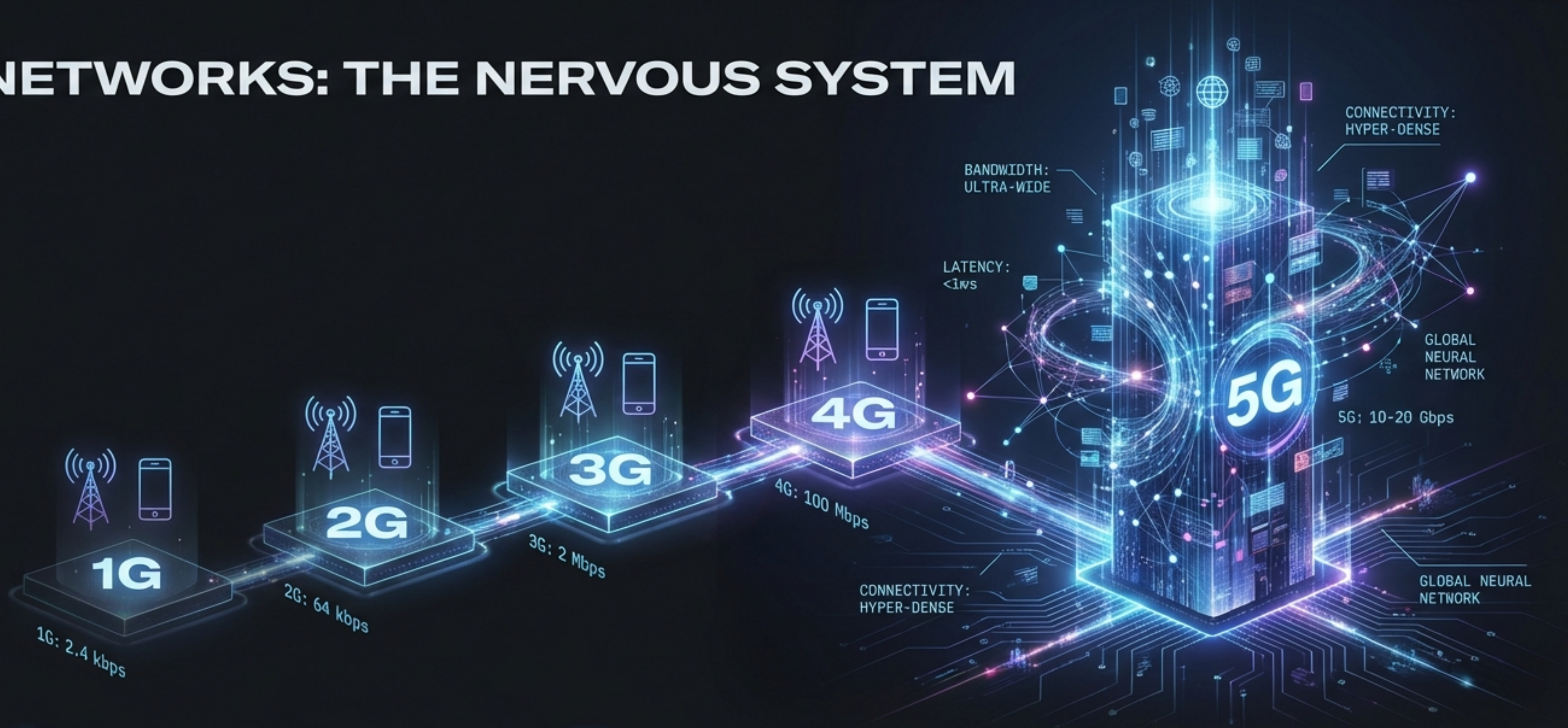
Generating news
articles and code.



INTEGRATING:

AlphaGo Zero learning
from scratch in 40
days to beat the world.

NETWORKS: THE NERVOUS SYSTEM



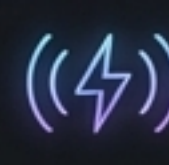
THE SPEED OF 5G:

Download a movie in the time it takes to read a sentence.



THE IMPACT:

4.2 Billion new minds coming online.



THE SHIFT:

Instantaneous global connection.

THE SPACE RACE FOR INTERNET



ONE PLANET. ONE PEOPLE.

SENSORS: THE ELECTRIC SKIN

OPTICAL HEART
RATE SENSOR (PPG)

SPACE MONO

ACCELEROMETER /
GYROSCOPE

SPACE MONO

MICROPROCESSOR

SPACE MONO

TEMPERATURE
SENSORS

SPACE MONO

BATTERY
CELL

EDA SENSOR

ACCELEROMETER

PULSE OXIMETER



CONVERGENCE:

SPACE MONO

Battery Size +
Sensor Sensitivity.



CAPABILITY:

SPACE MONO

Medical grade lab on a finger
(Heart rate 250x/second).



SCALE:

SPACE MONO

500 Billion connected
devices by 2030.



ECONOMY:

SPACE MONO

\$14.2 Trillion
IoT market.

2028: THE PACKET-SWITCHED LIFE

Cleveland to Wall Street: 59 Minutes



AI anticipates
schedule.



Sleeper vehicle
pickup.



Hyperloop
transit.



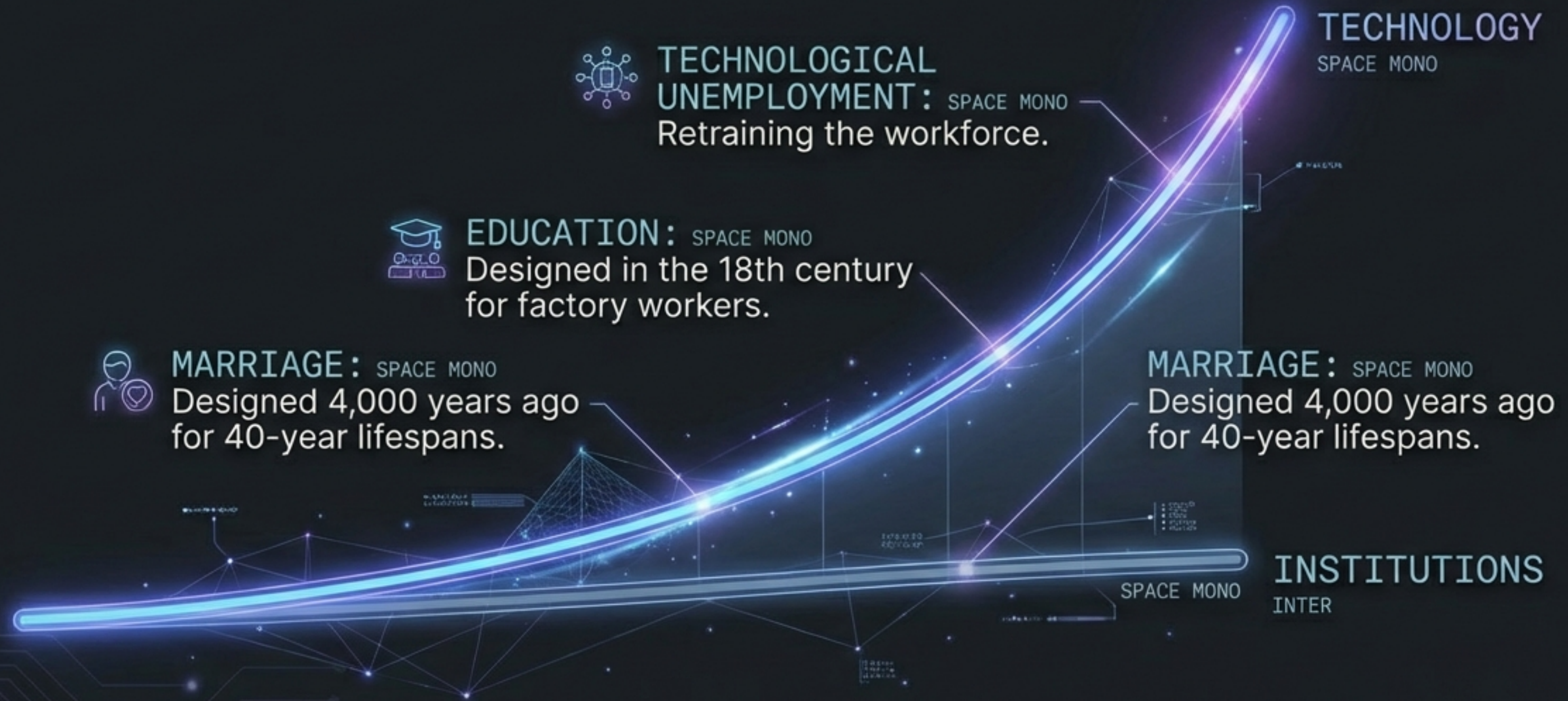
eVTOL to
rooftop.



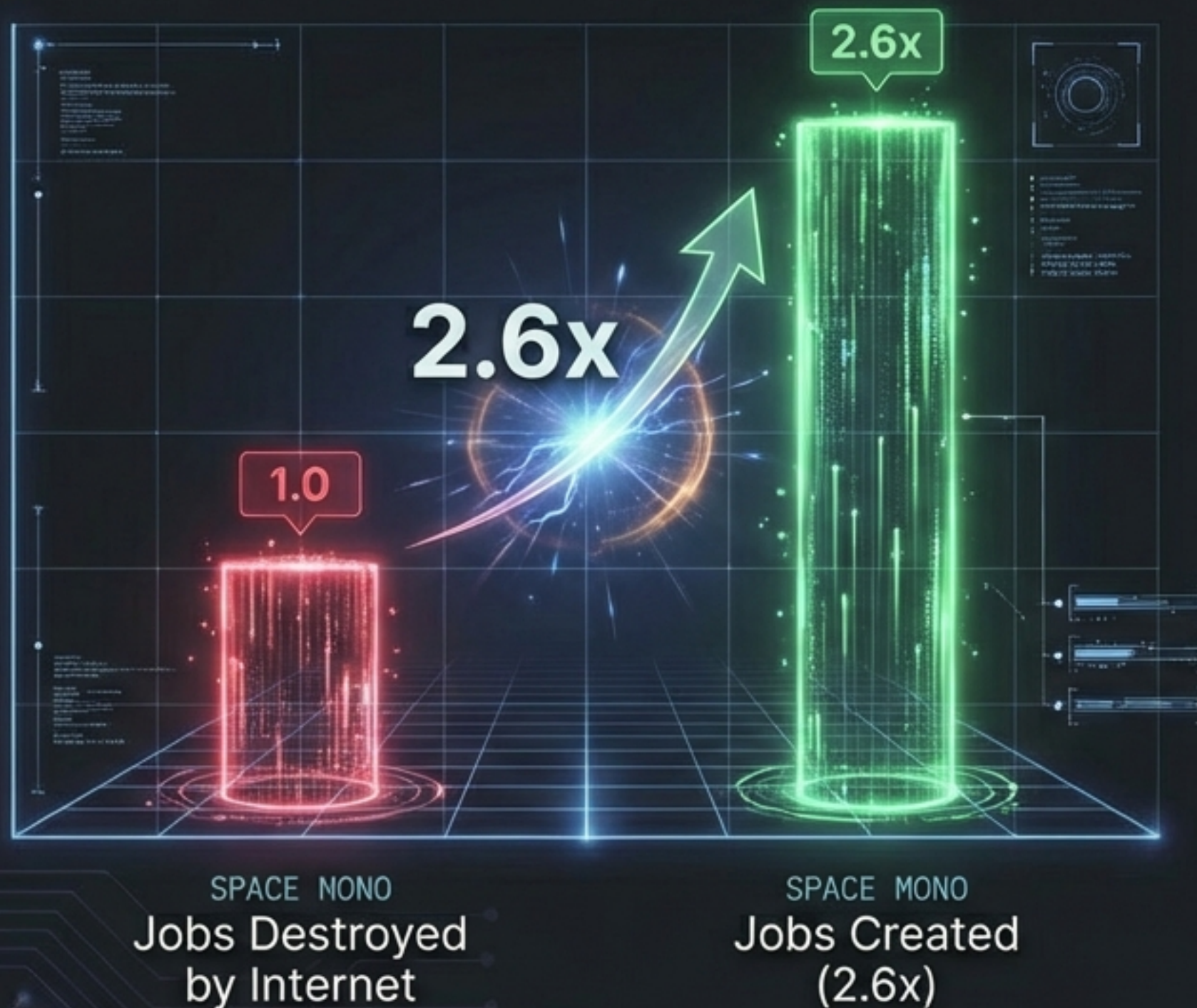
Arrival.

Seamless transitions. Priority choice: Speed, Comfort, or Cost.

THE CHALLENGE: INSTITUTIONAL LAG



THE OPPORTUNITY: ABUNDANCE



SPACE MONO

We are looking at a **century** of **wealth creation** compressed into the next decade.

SPACE MONO

40% of Fortune 500 companies will turn over in 10 years.

SPACE MONO

The time from Idea to Billion-Dollar Company is shrinking.

WELCOME TO AN ERA OF EXTRAORDINARY.

The Future Is Faster Than You Think

Space Mono

Space Mono

05.4.2023
508X DMH DMH